

PyTurbo_SF: An Adaptive Bootstrap Framework for Efficient Structure Function Analysis in Turbulent Flows

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Summary

Structure functions are fundamental diagnostic tools in turbulence research that reveal energy cascade characteristics, intermittency properties, and scaling laws by quantifying spatial correlations between field differences at varying separation distances (Frisch, 1995; Pope, 2000). PyTurbo_SF is a comprehensive Python package that tackles structure function analysis through an innovative adaptive bootstrap framework, enabling efficient and statistically rigorous calculations for 1D, 2D, and 3D turbulent datasets.

The package addresses critical computational bottlenecks in modern turbulence analysis by implementing power-of-2 spacing (sampling indices) strategies, adaptive convergence monitoring, and memory-optimized algorithms that automatically scale with dataset characteristics. PyTurbo_SF provides unprecedented statistical rigor through bootstrap resampling, delivering robust uncertainty quantification that has been missing from existing structure function tools. The software seamlessly processes xarray Datasets and supports diverse structure function types including longitudinal, transverse, scalar, advective, and energy flux functions recently developed for analyzing geophysical energy transfers (Pearson et al., 2021, 2024).

Applications span from oceanographic mooring time series and satellite altimetry measurements to high-resolution numerical simulations, enabling researchers to apply consistent, statistically sound methodology across scales from laboratory turbulence to planetary circulation systems.

Statement of need

Contemporary turbulence research increasingly relies on massive, high-resolution datasets from advanced observational platforms (satellite missions, autonomous platforms) and computational simulations (large eddy simulations, direct numerical simulations). Traditional structure function calculations face severe limitations: computational intractability for large 2D/3D datasets, absence of rigorous uncertainty quantification, manual parameter tuning requirements, and limited function types that restrict scientific insights.

Existing software tools address only subsets of these challenges. fastSF provides parallelized C++ implementations but lacks advanced function types and uncertainty quantification (Sadhukhan et al., 2021). MATLAB-based toolkits offer specific analyses but are environmentally limited and lack comprehensive statistical frameworks (Fuchs et al., 2022). Alternative approaches like coarse-graining (FlowSieve) address different aspects of turbulence analysis but cannot

40 provide the detailed scale-by-scale information that structure functions uniquely deliver (Storer
41 & Aluie, 2023).

42 The scientific community urgently needs tools capable of analyzing emerging datasets from
43 missions like NASA's SWOT satellite, which provides unprecedented high-resolution ocean
44 surface measurements, and next-generation atmospheric simulations that generate terabyte-
45 scale 3D datasets. Recent advances in structure function theory, particularly advective structure
46 functions for quantifying energy flux transfers (Pearson et al., 2021) and spectral flux estimation
47 methodologies (Pearson et al., 2024), require computational frameworks that can handle both
48 traditional and novel function types with statistical rigor.

49 PyTurbo_SF fills this critical gap by providing the first comprehensive, statistically robust
50 framework for structure function analysis that scales from small observational datasets to
51 massive simulation outputs while delivering quantified uncertainties essential for scientific
52 interpretation.

53 Software functionality

54 PyTurbo_SF implements the complete mathematical framework for structure function analysis,
55 supporting functions of the form $S_n(r) = \langle |\phi(\mathbf{x} + \mathbf{r}) - \phi(\mathbf{x})|^n \rangle_{\mathbf{x}}$ where ϕ represents arbitrary
56 field variables (velocity, scalars, derived quantities), $\mathbf{r}$ is the separation vector, n is the order,
57 and $\langle \cdot \rangle_{\mathbf{x}}$ denotes spatial averaging. The package supports traditional functions (longitudi-
58 nal, transverse, scalar), cross functions (longitudinal-transverse, scalar-scalar) and advective
59 structure functions that enable direct energy flux quantification (Pearson et al., 2021) and
60 pressure-work structure functions for analyzing energy transfer mechanisms.

61 The core algorithmic breakthrough is the adaptive bootstrap methodology that increases
62 computational efficiency and statistical reliability. The algorithm employs power-of-2 spacings
63 (1, 2, 4, 8, 16, ...) that optimize memory access patterns while providing optimal scale separation.
64 Adaptive convergence monitoring dynamically allocates computational resources by grouping
65 unconverged bins by characteristics and prioritizing bootstrap samples based on statistical
66 effectiveness. This approach eliminates manual parameter tuning while guaranteeing robust
67 uncertainty estimates through principled resampling.

68 Performance optimization enables analysis of previously intractable datasets. Memory-efficient
69 data structures and intelligent garbage collection maintain peak memory usage at 2-5× base
70 dataset size. Parallel processing via multiple backends (multiprocessing, threading, loky)
71 provides near-linear scaling with available cores. Benchmark testing with 2D DYCOMS
72 turbulence data demonstrates time complexity scaling of $O(NM \log N \log M)$ for 2D data,
73 enabling analysis of datasets with millions of grid points.

74 The package provides three main interfaces optimized for different dimensionalities:
75 `bin_sf_1d()` for oceanographic and atmospheric time series, `bin_sf_2d()` for satellite
76 and model surface fields, and `bin_sf_3d()` for volumetric simulation data. All functions
77 automatically detect data characteristics and optimize computational strategies accordingly,
78 while supporting both isotropic averaging and directional analysis for anisotropic flows.

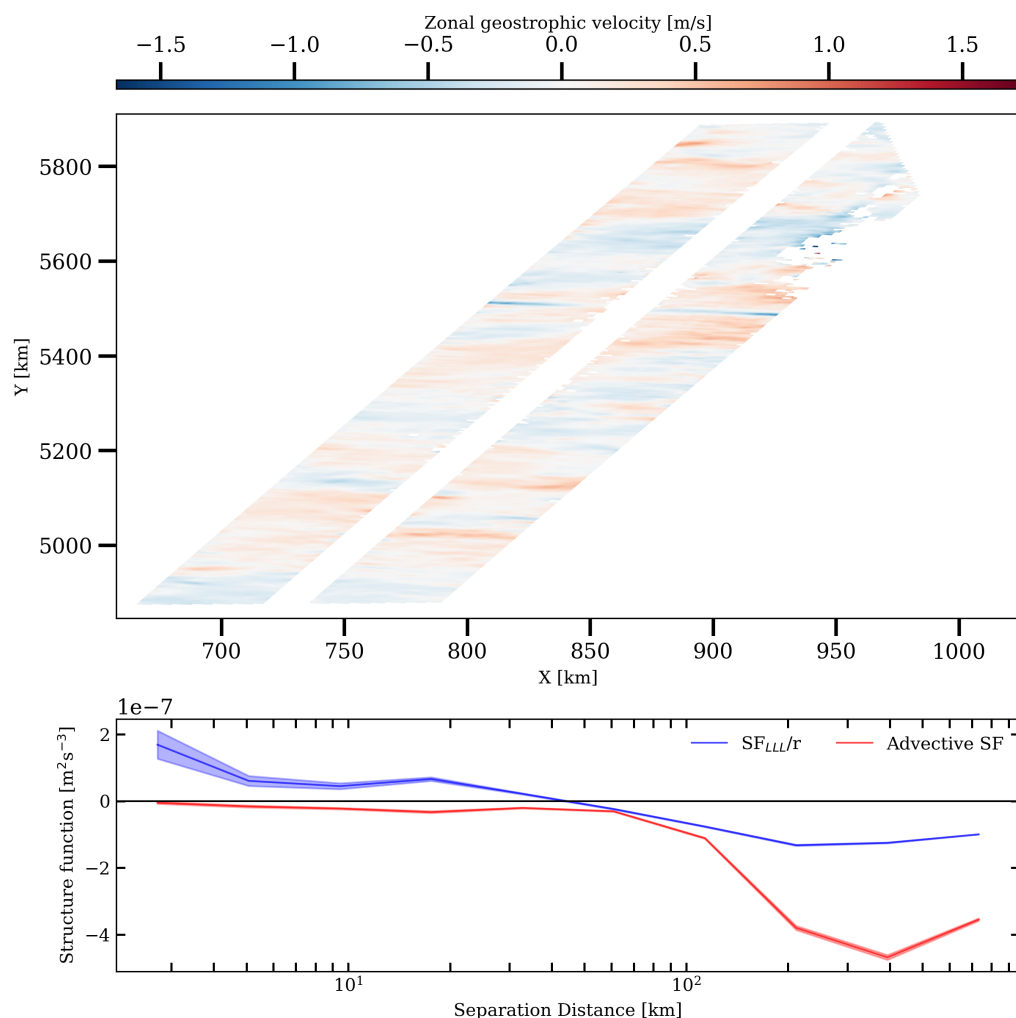


Figure 1: Satellite altimetry analysis showing second-order longitudinal structure functions calculated from SWOT data in the Gulf Stream region. Bootstrap error bars demonstrate statistical rigor while revealing energy cascade signatures in ocean surface turbulence. Results show the characteristic $r^{2/3}$ scaling predicted by geostrophic turbulence theory.

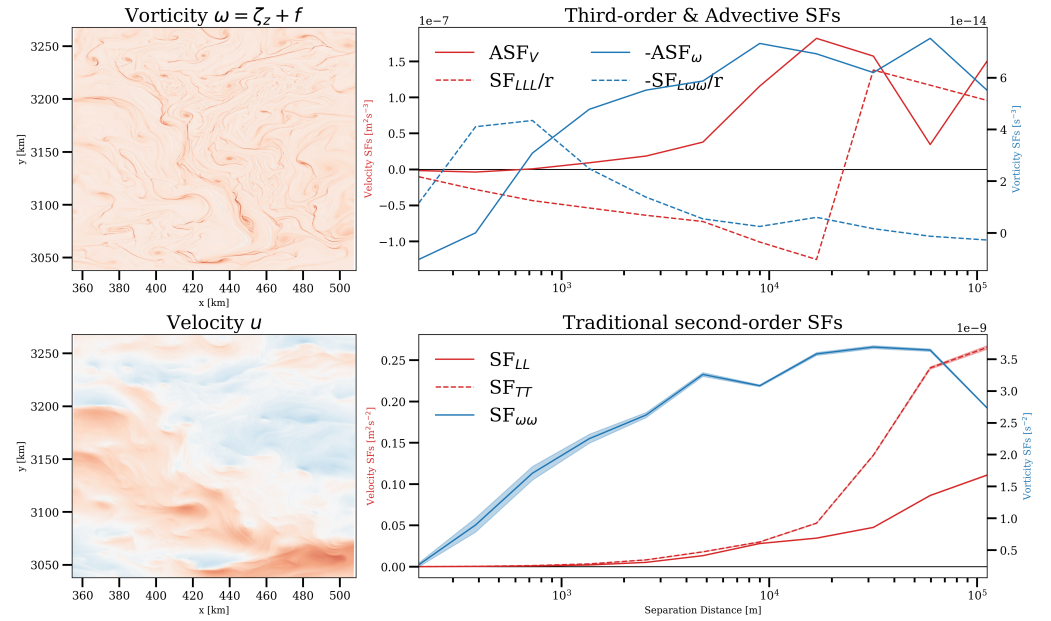


Figure 2: Regional ocean model analysis using CROCO simulation data showing energy transfer mechanisms through combined velocity-scalar structure functions. The adaptive bootstrap algorithm efficiently handles the large 2D spatial dataset while providing robust uncertainty quantification for energy flux estimates.

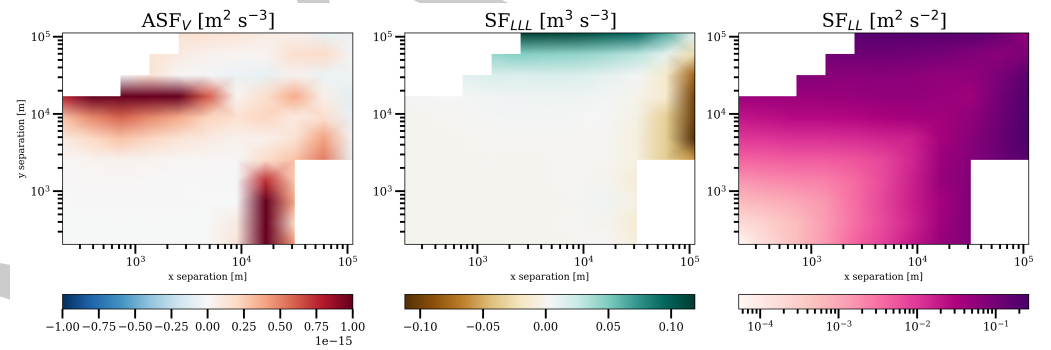


Figure 3: Regional ocean model analysis using CROCO simulation data showing energy transfer mechanisms through combined 2D velocity-scalar structure functions.

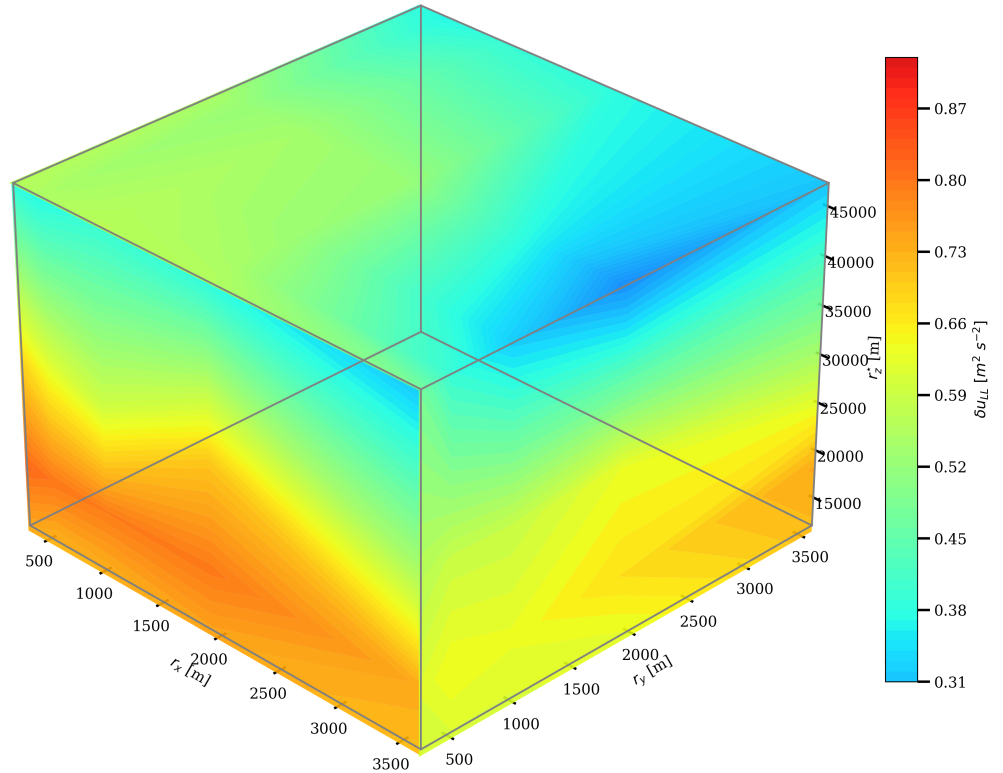


Figure 4: Atmospheric boundary layer turbulence analysis from DYCOMS large eddy simulation demonstrating 3D structure function capabilities. Volume-weighted binning accurately captures isotropic averaging in the 3D velocity field while bootstrap resampling provides statistical reliability for intermittency characterization.

Related work and scientific impact

PyTurbo_SF represents a significant advancement over existing structure function software by uniquely combining comprehensive function types, adaptive bootstrap methodology, and optimized computational algorithms. While fastSF provides basic parallelized calculations (Sadhukhan et al., 2021) and MATLAB toolkits offer specific analyses (Fuchs et al., 2022), no existing software delivers the combination of statistical rigor, computational efficiency, and methodological breadth required for contemporary turbulence research.

The package enables application of recent theoretical developments in structure function analysis, particularly advective structure functions that provide direct measurements of energy flux transfers in geophysical flows (Pearson et al., 2021). These functions reveal energy pathways that traditional approaches cannot capture, offering new insights into energy cascade mechanisms in ocean and atmospheric turbulence. The software also supports spectral flux estimation methodologies that bridge structure function analysis with spectral energy transfer studies (Pearson et al., 2024).

Scientific applications demonstrate the software's transformative impact across multiple domains. In oceanography, PyTurbo_SF enables analysis of satellite altimetry data for characterizing surface turbulence and energy dissipation rates. Atmospheric scientists can analyze large eddy simulation data to understand boundary layer turbulence and energy transfer mechanisms. The consistent methodology across platforms enables comparative studies that were previously

impossible due to software limitations and statistical uncertainties.

The adaptive bootstrap framework addresses a fundamental challenge in turbulence analysis: quantifying uncertainties in structure function estimates. Previous studies often reported results without rigorous error bars, limiting scientific interpretation and reproducibility. PyTurbo_SF's principled uncertainty quantification enables robust statistical comparisons and hypothesis testing, elevating the scientific standards of structure function analysis.

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